FitBuddy-AI Fitness Plan Generator Using Gemini Models

Government Arts and Science College-Manur-MSU142

TeamLeader:

Sandhiya P NM ID:F24D0CD882B6C7E9B9F44821FF74E70B

Team Members:Santhanamari G NM ID: C1A1D3256DDD23E87C0472B52EDBF20B

Saraswathi P NM ID: F7D8EF3617B74F0922A414AB1EDEF80E

Chinnathai S NM ID:6A593ADCC611910BA8240846346DA660

Muthumari M NM ID: 61F719397D001D3BE07494880A4743B7

Project Documentation Phase

Here's the *Project Documentation Phase* for FitBuddy – AI Fitness Plan Generator, which ensures the project is well-recorded, easy to maintain, and ready for stakeholder review:

📖 Project Documentation Phase: FitBuddy

1. Purpose of Documentation

- Provide a *comprehensive record* of the project's objectives, design, development, and testing.

- Serve as a *reference guide* for developers, trainers, and future contributors.

- Ensure *maintainability and scalability* by clearly outlining workflows and dependencies.

2. Types of Documentation

- *Technical Documentation*

- System architecture diagrams.

- API endpoints (/generate-workout, /submit-feedback, /view-all-users).
- Database schema (User, WorkoutPlan tables).
- Code structure (main.py, routes.py, gemini_generator.py, etc.).
- Deployment instructions (FastAPI + Uvicorn setup, environment variables).

- *User Documentation*
- Step-by-step guide for generating workout plans.
- Instructions for submitting feedback and receiving updated plans.
- Admin dashboard usage guide.
- Screenshots of UI pages (index.html, result.html, all_users.html).

- *Testing Documentation*
- Test case matrix (mapping requirements to test scenarios).
- Bug reports and resolution logs.
- Performance and security test results.

- *Project Management Documentation*
- Requirement analysis summary.
- Design phase artifacts (flowcharts, module descriptions).
- Planning phase (milestones, timelines, resource allocation).
- Development logs and version control notes.

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3. Documentation Tools & Formats

- *Markdown/HTML* → lightweight, developer-friendly documentation.
- *PDF Reports* → formal stakeholder presentations.
- *README.md* → quick-start guide for developers.

- *API Docs* → auto-generated via FastAPI /docs.

4. Deliverables

- *System Requirement Specification (SRS)* document.

- *Design Document* (architecture + module descriptions).

- *Developer Guide* (setup, code structure, API usage).

- *User Manual* (how to use FitBuddy).

- *Test Report* (results of unit, integration, and UAT testing).

- *Deployment Guide* (local setup + cloud readiness).

5. Benefits

- Ensures *knowledge transfer* for new developers.

- Provides *clarity for stakeholders* on project scope and outcomes.

- Facilitates *future upgrades* (cloud deployment, wearable integration).

- Acts as a *compliance record* for project lifecycle.